Exercise 2.4 Universal Property of Affine Scheme

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Problem (ex.2.4)**.** Let A be a ring and $(X, \mathcal{O}_X)$ be a scheme. Then show that there is a natural bijection

$$\mathrm{Hom}_{\mathsf{Sch}}(X, \mathrm{Spec}(A)) \xrightarrow{1:1} \mathrm{Hom}_{\mathsf{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

In fact, we only used X is a *locally ringed space* in the proof. We'll actually prove that

$$\mathrm{Hom}_{\mathsf{LRS}}(X, \mathrm{Spec}(A)) \xrightarrow{1:1} \mathrm{Hom}_{\mathsf{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Construction of forward arrow. From a morphism $f : X \to \mathrm{Spec}(A)$ of locally ringed spaces, we have a sheaf morphism $f^\# : \mathcal{O}_{\mathrm{Spec}(A)} \to f_*\mathcal{O}_X$. Taking global sections we get a ring homomorphism $A \to \Gamma(X, \mathcal{O}_X)$.

Construction of backward arrow. Let $\phi : A \to \Gamma(X, \mathcal{O}_X)$ be a ring homomorphism. We define the continuous map $f : X \to \mathrm{Spec}(A)$ as follows. For $x \in X$, we consider the composition ϕ_x of ϕ with the germ map $\Gamma(X, \mathcal{O}_X) \to \mathcal{O}_{X,x}$. Let

$$f(x) := \phi_x^{-1}(\mathfrak{m}_x).$$

Next, we need to define a sheaf map $f^\# : \mathcal{O}_{\mathrm{Spec}(A)} \to f_*\mathcal{O}_X$. By general knowledge on sheaves on a base (see https://stacks.math.columbia.edu/tag/009O), it suffices to define the map for the distinguished opens $D(\alpha)$ and check they agree on intersections (by checking on smaller distinguished opens contained in the intersections).

For $\alpha \in A$, the desired homomorphism $f^\#(D(\alpha)) : A_\alpha \to \Gamma(f^{-1}(D(\alpha)), \mathcal{O}_X)$ is naturally induced from ϕ via localization. Indeed, on the open set

$$f^{-1}(D(\alpha)) = \{x \in X : \alpha \notin \phi_x^{-1}(\mathfrak{m}_x)\} = \{x \in X : \phi(\alpha)(x) \neq 0\},$$

the function $\phi(\alpha)$ is invertible. (This does not use any property of a scheme, only requiring X to be a LRS. It follows from the definition of a local ring, $\mathfrak{m} = R \setminus R^\times$, that every non-vanishing function is locally invertible. Since the inverses are unique, they patch together, so every non-vanishing function has a global inverse.)

We must also check that $(f, f^\#) : X \to \operatorname{Spec}(A)$ is a morphism of locally ringed spaces. We need to check that, for every $x \in X$, the map

$$f_x^\# : \mathcal{O}_{\operatorname{Spec}(A), f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is a local homomorphism. That is, $f_x^\#(\mathfrak{m}_{f(x)}) \subset \mathfrak{m}_x$. This is clear from the definition of f and $f^\#$, since

$$f(x) = \phi_x^{-1}(\mathfrak{m}_x) = \{\alpha \in A : \phi(\alpha)_x \in \mathfrak{m}_x\}.$$

Check backward $\circ$ forward is id. We are given $f : X \to \operatorname{Spec}(A)$. First we need to check that $\phi_x^{-1}(\mathfrak{m}_x) = f(x)$.

We have the commutative diagram

$$\begin{tikzcd}
\Gamma(\operatorname{Spec}(A), \mathcal{O}_{\operatorname{Spec}(A)}) \arrow[r] \arrow[d] & \Gamma(X, \mathcal{O}_X) \arrow[d] \\
\mathcal{O}_{\operatorname{Spec}(A), f(x)} \arrow[r, "f_x^\#"] & \mathcal{O}_{X,x}
\end{tikzcd}$$

where the composition along is by definition ϕ_x. So we can compute $\phi_x^{-1}(\mathfrak{m}_x)$ along the direction. Since $f_x^\#$ is a local homomorphism, we get

$$\begin{tikzcd}
f(x) & \\
\mathfrak{m}_{f(x)} \arrow[u, squiggly] & \mathfrak{m}_x \arrow[l, squiggly]
\end{tikzcd}$$

as desired.

The sheaf morphism we constructed is clearly unique.

Check forward $\circ$ backward is id. This is trivial.

Solution. This completes the proof. $\square$